

PCBoard 15.4 / 15.41

Addendum to the PCBoard Manual

This document supplements the PCBoard printed manual (547 pages). It covers all changes from version 15.3 through 15.41.

PCBoard 15.4: Clark Development Company (unreleased beta)

PCBoard 15.41: pcbirc crew, August 2026

github.com/verta1878/pcbirc

Clark's original source code is proprietary.

pcbirc crew additions are GPLv3.

CodeBase library is LGPL v3.0.

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1. PCBoard 15.4 Changes

PCBoard 15.4 was never officially released by Clark Development Company. These changes were found in the 15.4 beta source code and MKPCBTXT.EXE binary. This section reconstructs what Clark intended for 15.4 based on source code analysis.

1.1 Personal Storage Area (PSA_PERSONAL)

PCBoard 15.4 adds user profile fields as a new Personal Storage Area (PSA):

- **Gender (M/F)**
- **Birthdate**
- **WEB address (URL)**

These fields are stored in the user record as PSA_PERSONAL, a new entry in PCBoard's modular PSA system. The complete list of PSA types:

PSA Type	Purpose
PSA_ALIAS	User aliases
PSA_VERIFY	Verification data
PSA_ADDRESS	Mailing address
PSA_PASSWORD	Password history
PSA_STATS	Usage statistics
PSA_NOTES	Sysop notes
PSA_ACCOUNT	Account and billing
PSA_QWKNET	QWK networking
PSA_PERSONAL	Gender, birthdate, web (NEW in 15.4)
PSA_BANK	Time bank

The new fields are prompted during new user registration and profile updates. Sysops can customize the prompts by editing PCBTXT strings 747-750 (see next section).

1.2 PCBTEXT Strings 747-750

Four new PCBTEXT strings were added in version 15.4. The version tag in STRS15.C for these strings is 154, confirming they were intended for PCBoard 15.4.

Slot	Default Text	Purpose
747	Enter your gender (M/F)	PSA_PERSONAL gender
748	Enter your birthdate	PSA_PERSONAL birthdate
749	Enter your WEB address	PSA_PERSONAL URL
750	Enter color (B)lue,(G)reen,...	User color preference

Run MKPCBTXT to regenerate the PCBTEXT file after upgrading. These strings can be customized with any text editor.

PCBOARD.DAT is unchanged from 15.3. Line 349 remains the last field. No new PCBOARD.DAT lines were added in 15.4.

2. PCBoard 15.41 Changes

PCBoard 15.41 is the pcbirc crew's continuation of the 15.4 beta. It adds new utilities, fixes 26 bugs, and changes the compiler from Borland C++ 3.1 to OpenWatcom 2.0.

2.1 FilebaseFlags (CNAMES.ADD Extension)

The 64-byte Reserved field in each CNAMES.ADD conference record has been redefined as FilebaseFlags[64]. This provides per-filebase bit flags with one bit per filebase, supporting up to 512 filebases per conference.

Defined flags:

- Bit 0: FBFLAG_ISO -- marks a filebase as ISO/CD-ROM backed
- Bits 1-7: Reserved for future use

PCBoard 15.4 zero-initializes these bytes and never reads them. PCBSM skips them during conference operations. All existing 15.4 installations are forward-compatible with this change. No upgrade utility is needed.

2.2 PCBISO -- ISO/CD-ROM File Area Indexer

PCBISO is a standalone utility for managing ISO/CD-ROM file areas in PCBoard. It reads CNAMES.@@@ and CNAMES.ADD to find configured file areas, scans source paths, and generates PCBoard DIR listing text files with automatic FILE_ID.DIZ extraction from ZIP archives.

Commands:

Command	Description
PCBISO /INDEX path dirfile	Scan path, generate DIR listing
PCBISO /INDEX ALL cnames add	Index all ISO-backed conferences
PCBISO /REBUILD path dirfile	Delete and regenerate DIR listing
PCBISO /LIST cnames add	List configured file areas
PCBISO /STATUS cnames add	Detailed status with file counts
PCBISO /SETISO conf fb add	Mark filebase as ISO-backed
PCBISO /CLEARISO conf fb add	Clear ISO flag
PCBISO /MOUNT iso	Register ISO in mount table
PCBISO /UNMOUNT drive	Release drive letter

Example -- Index a CD-ROM:

```
PCBISO /SETISO 5 1 C:\PCB\GEN\CNAMES.ADD  
PCBISO /INDEX D:\ C:\PCB\GEN\DIR5
```

Example -- Index all ISO-backed areas:

```
PCBISO /INDEX ALL C:\PCB\GEN\CNAMES.@@@ C:\PCB\GEN\CNAMES.ADD
```

PCBISO uses the FilebaseFlags field in CNAMES.ADD to identify which file areas are ISO/CD-ROM backed. Set the flag with /SETISO before indexing.

Location: PCBSRC/MAIN/SOURCE/UTIL/PCBISO/

Compiles under GCC and OpenWatcom 2.0 (42K DOS4G binary).

2.3 QFront -- FidoNet Mailer Suite

QFront is a clean-room FidoNet front-end mailer and mail processor for PCBoard 15.3 and 15.4. It replaces external mailers (BinkleyTerm, FrontDoor) for FidoNet connectivity.

Programs included:

Program	Description	Lines
QFRONT.EXE	Main mailer and call handler	889
QFCONFIG.EXE	TUI configuration editor	720
QSCAN.EXE	EchoMail tosser and scanner	848
QNLIST.EXE	Nodelist compiler	624
QFUTIL.EXE	Command-line utility	162

Total: 21 source files, 10,123 lines, GPLv3. Supports EMSI, YooHoo, and FTS-1 handshake protocols with Zmodem and Xmodem file transfer.

See tools/qfront/QFRONT.DOC for the complete QFront manual (38 pages) and tools/qfront/INSTALL.TXT for PCBoard integration instructions.

BOARD.BAT Integration:

```
: LOOP
QFRONT.EXE qfront.cfg
IF ERRORLEVEL 5 GOTO FAX
IF ERRORLEVEL 1 GOTO BBS
GOTO LOOP
: BBS
PCBOARD.EXE
GOTO LOOP
```

2.4 Compiler Change (OpenWatcom 2.0)

PCBoard 15.41 is compiled with OpenWatcom 2.0 (openwatcom2irc fork, r0.6.0) replacing Borland C++ 3.1. All 28 Clark binaries target DOS4G protected mode (32-bit).

Borland-specific code (inline ASM, conio.h, dos.h) has been replaced with Watcom-compatible equivalents. The openwatcom2irc fork adds 5 targets to the standard OpenWatcom distribution.

See BUILD.md in the source tree for detailed instructions.

3. Bug Fixes (26 total)

3.1 pcbis Bugs (11)

No.	Description
1	Crash on empty dir list -- null pointer dereference
2	Double-close on file handle after error path
3	Missing bounds check on conference number
4	Garbled display on 80-column terminal overflow
5	Off-by-one in group name copy (missing null terminator)
6	File count overflow at 65535 entries (uint16 wrap)
7	Sort comparator returns wrong sign for equal keys
8	Rebuild skips first file in directory listing
9	Date parsing fails on Y2K dates (2-digit year)
10	Temp file left behind on Ctrl-C during index build
11	Verbose mode crashes if log file path is too long

3.2 pcbfoss Bugs (6)

No.	Description
1	FOSSIL init (INT 14h/04) not checking return code 0x1954
2	DTR drop timing too short for some modems (under 500ms)
3	16550 FIFO detection clobbers IER register on non-16550
4	Baud rate divisor calculation wrong for 115200 (off by 1)
5	RTS/CTS flow control not disabled before Zmodem transfer
6	ser_close() doesn't restore original UART interrupt mask

3.3 QFront Bugs (9)

No.	Description
1	No signal handler -- Ctrl-C left COM ports open, locks orphaned
2	No SIGPIPE handling -- broken pipe crash on serial write
3	BSO scan didn't handle .PNT subdirectories for points
4	Flow file parse didn't handle ^ and # directives
5	Zmodem CRC-32 table endianness issue on big-endian systems
6	EMSI timeout wasn't applied to DAT packet receive
7	Nodelist parser choked on Boss keyword (pointlists)
8	.BSY lock used fopen() -- TOCTOU race with other nodes
9	QNLIST had no rollback on CRC failure after nodediff apply

4. Developer Reference

4.1 PCBOARD.DAT Format

PCBOARD.DAT is a plain TEXT file with one field per line. Clark appended new lines between versions (15.0 had fewer lines than 15.3). The current format has 349 lines.

PCBoard 15.41 does NOT add new PCBOARD.DAT lines. Line 349 remains the last field. Future extensions will append lines after 349 following Clark's established pattern.

See docs/devkit/PCBDAT.DOC for the complete field-by-field specification.

4.2 CNAMES.ADD Format Changes

Each CNAMES.ADD record is 256 bytes (addconftype in NEWDATA.H). The change in 15.41 renames bytes 192-255 from Reserved[64] to FilebaseFlags[64].

CNAMES.ADD record layout (256 bytes):

Offset	Size	Field	Description
0	32	EchoTag	EchoMail area tag
32	68	FidoAddr	FidoNet address string
100	4	ReqSecLevel	Required security level
104	24	ReqSecFlags	Required security flags
128	64	Various	Conference-specific fields
192	64	FilebaseFlags	Per-filebase bit flags (was Reserved)

4.3 Developer Kit Files

Clark's original developer kit (Develop.zip) contains 23 file format specification documents. These have been extracted to docs/devkit/:

File	Contents
PCBDAT.DOC	PCBOARD.DAT file format (text, one field per line)
CNAMES.DOC	CNAMES.@@@ and CNAMES.ADD conference formats
USERS.DOC	Users file format and record layout
MSG.S.DOC	Message base format (headers and index)
DIR.DOC	DIR listing text file format
STRUCTS.DOC	General data structures (C header equivalents)
DOORS.DOC	Door interface (PCBOARD.SYS, USERS.SYS)
CALLER.DOC	Caller log format
SCRIPT.DOC	Script/questionnaire file format
PPE.DOC	PPL compiled .PPE file format

Plus 13 additional format specifications. See docs/devkit/README.1ST for the full list.

5. Building from Source

PCBoard 15.41 builds with the openwatcom2irc fork of OpenWatcom 2.0 (release r0.6.0). The fork is at github.com/verta1878/openwatcom2irc.

Requirements:

- OpenWatcom 2.0 (openwatcom2irc r0.6.0)
- DOS4G/W extender (included with OpenWatcom)
- CodeBase 6.4 library (LGPL v3.0, from GitHub)

Quick build (DOS4G target):

```
set WATCOM=C:\WATCOM
set PATH=%WATCOM%\BINW;%PATH%
set INCLUDE=%WATCOM%\H
wcl386 /l=dos4g pcboard.c ...
```

QFront (Linux/GCC):

```
cd tools/qfront/src
gcc -o qfront -DQFRONT_LIB bso.c emsi.c events.c ...
```

See BUILD.md in the source tree for detailed instructions.

PCBoard 15.4/15.41 Addendum -- pcbirc crew 2026

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